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When Diagenesis Meets Geomechanics: An Example from a Deeply Buried Tight Sandstone Reservoir

Fahd Almalki, Andrew Rees, Francis Elisabeth, and Mohammad Mohanna, Unconventional Exploration and Characterization Department, Saudi Aramco, Saudi Arabia

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Abstract

Deeply buried sandstones go through significant diagenetic phases that evolve into tight reservoirs. Successful hydrocarbon production from these sandstones requires drilling long lateral wells and conducting robust geomechanical analysis to guide hydraulic fracturing stimulation. Yet, geomechanics and diagenesis are rarely linked together in reservoir studies. This paper explores the relationship between the geomechanical and diagenetic properties of a deeply buried tight sandstone reservoir.

A total of 973 core samples from 28 wells penetrating the target horizon were selected for thin section preparation for detailed petrographic analysis. Point-counting has been performed on each sample to extract semi-quantitative data to characterize the reservoir composition and diagenetic history. X-Ray Diffraction (XRD) and Scanning Electron Microscope (SEM) imaging were performed to identify the mineralogy and discern different clay types. Meanwhile, a full geomechanical assessment was conducted, calibrated by extensive geomechanical testing (Triaxial, multistage, micro-rebound and scratch tests) to measure the rock strength and to predict the subsurface stress conditions.

The tight sandstones had undergone an extensive level of compaction, demonstrated by welded grain contacts. Mineralogical alteration such as feldspar dissolution took place synchronous with quartz overgrowth cementation and fibrous to leaf-like illite clay mineral crystallization. A positive relationship exists between authigenic illite and porosity confirming the dominance of micro-porosity in the reservoir and also highlighting the reason for the low permeability due to small pore throats and very tortuous flow paths. The Unconfined Compressive Strength (UCS) from the geomechanical analysis showed a large variation with values classifying the sandstone from a strong to very strong rock. The UCS showed a clear negative relationship with porosity and a positive relationship with the percentage of quartz cementation in the set of samples. Therefore, reservoir diagenesis clearly influenced the rock porosity, permeability and its geomechanical characteristics.

Integrating various aspects of reservoir geosciences provides a robust understanding of the controlling factors that govern storage, flow capacity and fraccability of tight sandstone reservoirs. The linkage between reservoir diagenesis and geomechanics demonstrates a well-founded integration and characterization of the reservoir's macro to micro scale features. Hence, these relationships can be incorporated into the static and

dynamic reservoir models for a more accurate prediction of the field production and for improved decision making.

Introduction

Tight sandstone reservoirs are generally characterized by low permeability resulting in low well productivity. The low permeability is a result of the reservoir's complex reservoir characteristics particularly, deep burial, wide distribution and complex depositional and diagenetic processes (Song et al., 2024). However, recent advances in the drilling and stimulation technologies have enabled the liberation of significant hydrocarbon resources from the subsurface for commercial production. Detailed analysis of the macro and microscopic features of tight sandstones is essential to generate data-driven interpretations that are transformed into inputs for reservoir characterization models and to support well planning, drilling and hydraulic fracture stimulation.

In this study, we characterize a deeply buried tight sandstone reservoir that has been through a long history of diagenetic evolution causing a reduction of porosity and permeability yet preserved a significant network of inter-connected micro-pore system. Then we share the relevant geomechanical characteristics of the reservoir that have been documented to be primarily controlled by the level and type of diagenetic minerals that are formed authigenically in the pore system.

Methodology

In order to achieve a robust understanding of a tight sandstone reservoir, an intensive program of well drilling, petrophysical wireline logging and core samples acquisition is required in order to perform a range of routine and special core analyses including geomechanical testing. The main difference between conventional sandstones reservoirs with high porosity and permeability and tight sandstones is in the level of post-depositional processes that often mask the original controls in reservoir quality leading to a much narrower range in the pore volume whilst reducing the permeability several orders of magnitudes. In clastic sedimentary reservoirs, the original sandstone texture such as grain size, sorting, sedimentary structures and the mineralogical composition define a set of facies that are deposited under specific depositional processes and environments generating subsurface reservoirs with a wide range of porosity (0 to 33%). On the other hand, tight sandstones usually go through a long diagenetic evolution that can destroy or preserve pore volume and its connectivity. One of the most common mechanism for porosity destruction is compaction and cementation, whilst certain processes such as grain dissolution labile grains in addition to grain coating and micro-fibrous clay mineral growth may lead to porosity enhancement or preservation (Figure 1).



Figure 1—Schematic diagram of the range of porosities seen in conventional sandstone reservoir vs. tight sandstones with the primary controls in reservoir quality manifested by original depositional and post-depositional processes.

In this study, a total of 28 wells were selected for a thorough sedimentological and diagenetic analysis to unravel the original depositional and post-depositional characteristics of the reservoir. Most of these wells were cored with a total core footage of 2,900 ft of cores, whilst a few of them had only side wall cores (SWC). All cores are plugged horizontally for conventional core analysis (CCA) generating porosity and permeability data. The plugging is done at a consistent footage, whenever core condition allows, with no biases to certain facies or rock types to generate a statistically representative distribution of the reservoir petrophysical properties. The cores were described fully focusing on the lithology, grain textural variability such as grain size trends and level of grain sorting. Bed boundaries, sedimentary structures, presence or absence of bioturbation, hydrocarbon staining and fractures have been documented. The core description are integrated with CCA plug data and a basic set of petrophysical wireline logs data such GR, Neutro-density and sonic travel time in comprehensive core description charts.

973 plugs were selected for detailed quantitative petrographic and diagenetic study at approximately a constant sampling rate of 33% (i.e. 1 sample per 3 core plugs). This study consisted of thin section slides preparation, thin section photography, and a visual description of the grain size, roundness and sorting. The mineralogic composition was also identified in addition to a documentation of the cement type and pore condition. The visual petrographic description was complemented by extremely valuable and useful quantitative data generated through point-count modal analysis at 300 points per each thin section. The 300-point count data provided rich insights and data-driven interpretations to unravel the main controls on the reservoir quality and in particular to the understanding of pore types and their contribution to the overall plug porosity measured by CCA analysis.

A subset of 143 samples was selected for X-Ray Diffraction (XRD) analysis and Scanning Electron Microscope (SEM) imaging. XRD provides a quantitative breakdown of the mineralogy of very small crushed samples (20 to 30 mg in weight). The XRD analysis has been conducted on whole core (WH) samples as well as the clay fraction (CF) representing the fine material that are less than 2 μ m in grain size to quantify the clay minerals present in the samples. SEM imaging was conducted to provide a high-resolution visualization of the pore network and crystalline fabric of the clay mineralogy within the set of chosen samples (Figure 2).

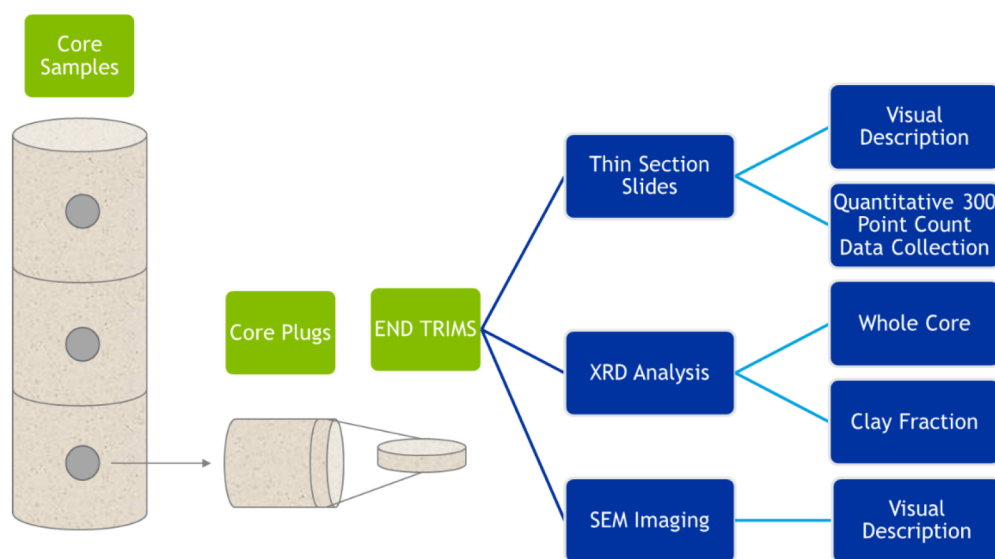


Figure 2—Workflow chart to illustrate the methodology followed for reservoir sedimentological, mineralogical and diagenetic characterization whilst acquiring quantitative point count data and SEM imaging.

Tight sandstones are known to exhibit extensive variation in strength (Higgs, 1981; Plumb, 1994). This variability is attributed to factors such as mineral composition, porosity, and cementation (Pittman, 1979;

Nelson, 2001). To characterize the strength of our sandstone, a comprehensive laboratory testing campaign was conducted across 18 wells in 7 fields. The testing program included:

- 73 Uniaxial Compressive Strength (UCS) tests to determine the rock's strength under unconfined conditions (ASTM, 2014)
- 93 Triaxial tests to determine the elastic properties and friction angle under various confining pressures, simulating downhole in-situ conditions (ISRM, 2007)
- 1107 Micro Rebound Tests (MRT) using a Rebound Hammer to estimate the rock's surface hardness and derive a continuous static Young's modulus and UCS profile (Katz et al., 2000)
- 398 UCS points from Scratch tests to create an indentation along the full core or plugs and derive a continuous UCS profile at a millimeter scale (Huang et al., 2011)

To compute the full profile of Uniaxial Compressive Strength (UCS), a traditional approach is employed, correlating UCS with petrophysical logs, such as porosity and velocity, or with elasticity parameters, like Young's modulus (Pickett, 1963; Han et al., 1986). The workflow is illustrated in Figure 3.

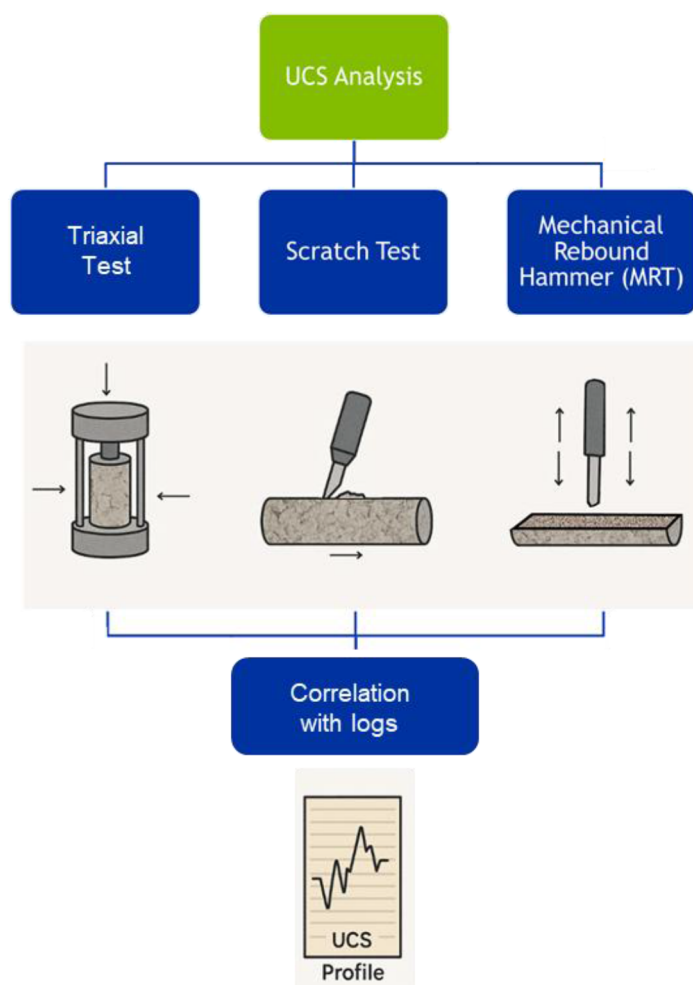


Figure 3—Workflow chart to illustrate the methodology followed for geomechanical testing on core samples and then the correlation and calibration with logs to generate a continuous UCS profile for the complete reservoir section.

Results

Sedimentological and Diagenetic Characterization

Fine to medium grained, thickly bedded and blocky sandstone was identified through core observation and description. The succession in many wells consists of very high net over gross with very thin, discontinuous, argillaceous sandstone and siltstone layers that are sandwiched between much thicker sandy beds. In terms of sedimentary structure, the sandstones appear to be massive, with some cryptic cross bedding with abundant signs of dewatering such as dish structure and dewatering pipes. In some parts, the predominant sedimentary structure is flat upper plane beds with no obvious bed boundaries and contacts (Figure 4).



Figure 4—A representative core photo of one of the wells of the study showing the typical sandstone dominated lithology of the tight reservoir with no obvious bed boundaries suggesting high deposition rate with very high sediment supply generating amalgamated and blocky sandstones.

The grain texture is frequently composed of sub-rounded to sub-angular, poorly to moderately sorted and in some cases shows a bimodal sorting composed of medium to coarse grains that are embedded in a matrix of fine sand grains (Figure 5).

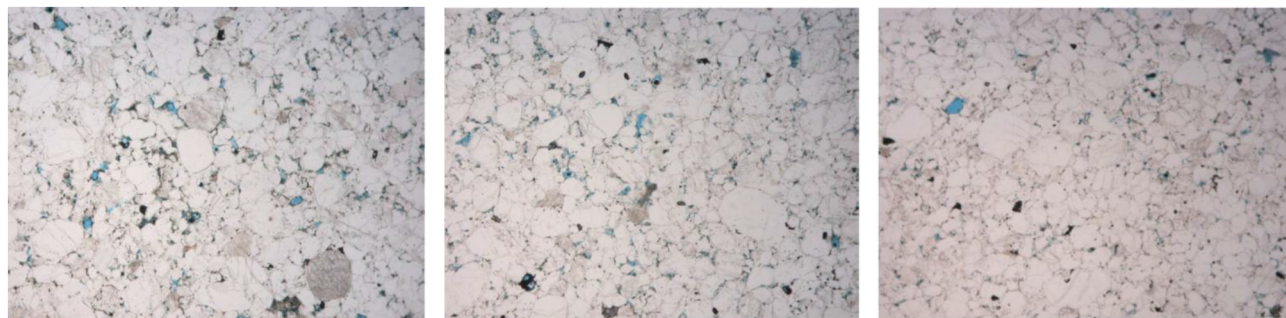


Figure 5—Representative core photos showing the moderate to pore sorting of the sandstone grains, tight packing and perhaps high level of cementation.

Overall the sandstone shows a quartz arenitic lithology with very high monocrystalline quartz seen in the thin sections. This has also been confirmed with XRD analysis showing an average quartz composition at 89.0%, followed by plagioclase feldspar 3.5%, then k-feldspar 2.5%, dolomite 1.9%, calcite 1.0%, illite/mica 1.5% and some rare sulphide minerals such as pyrite and baryte (Figure 6).

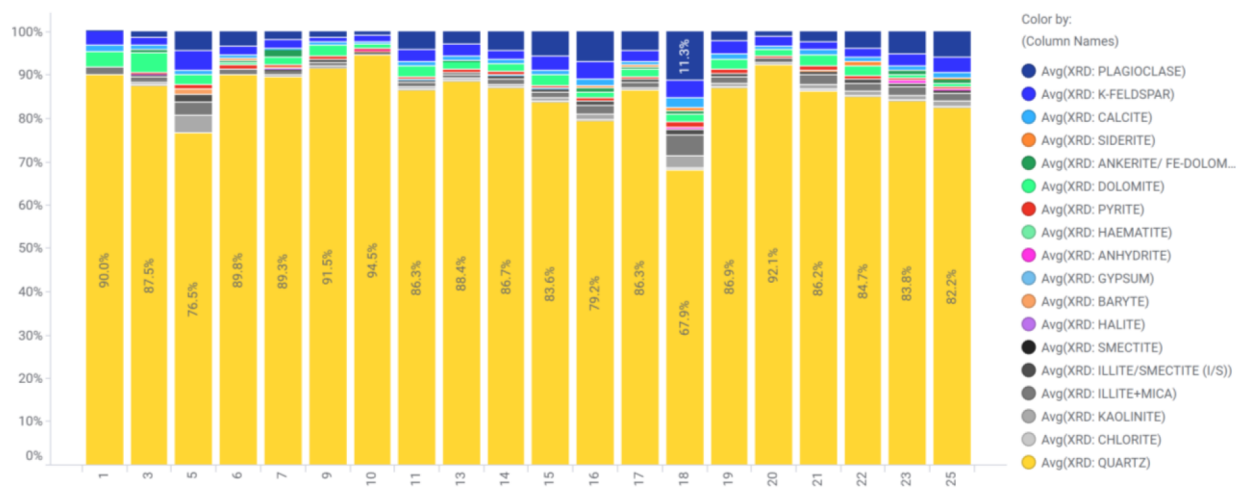


Figure 6—Bar chart showing the whole core average mineralogy percentages from samples in each well showing the high quartz content in the tight sandstone reservoir (X-axis is well number)

XRD data on the clay fraction was done generating a high resolution breakdown of the clay minerals composition. Illite clay minerals is the dominant clay mineral measured by XRD in almost all of the samples averaging 62.2%, followed by mixed illite/smectite at 10.3% whilst chlorite and kaolinite measured 15.1% and 12.4%, respectively. The sandstone appears to be completely devoid of any smectite clay minerals in all analyzed samples (Figure 7).

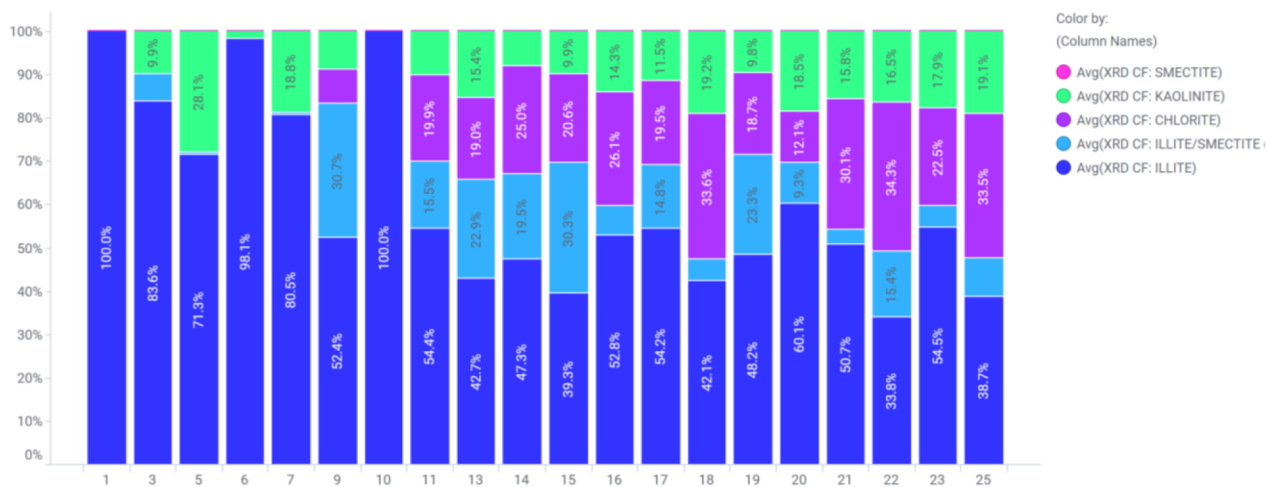


Figure 7—Bar chart showing the clay fraction average mineralogy percentages in each well showing the dominance of illite followed by chlorite and lastly kaolinite clay minerals (X-axis is well number)

Whilst XRD data can provide quantitative mineralogy analysis, it cannot distinguish grains from cement which limits the ability of XRD data to understand diagenetic alterations to the original sediments. However, petrographic investigation coupled with the 300-point count modal analysis illuminated the level of compaction, cementation and pore types in the tight sandstones. The results show that the sandstones have been subjected to a moderate to high level of compaction manifested by re-arrangement of grains, inter-granular pressure dissolution and grain suturing. Grain welding is commonly seen in argillaceous sandstones and in some cases show stylolitization. Grain compaction has a considerable influence in reducing the inter-granular volume to much less than the porosity of the original and pre-compaction sediment. In terms of cementation, the results show that the most common types of cements are quartz followed by authigenic illite clay. The latter agree very well with the XRD clay fraction data. Classifying samples by the level

of cementation at 10% cutoff to determine the most dominant cement, whilst excluding the argillaceous sandstones with high detrital clay material. The data show that 69.5% of the samples are dominated by quartz cement, followed by 15.8% with authigenic illite cement, 10.7% is a mixture of quartz and illite at almost equal proportions in every sample. The remaining 4% consists of various dolomite, calcite, and baryte seen as rare and accessory cementing minerals (Figure 8).

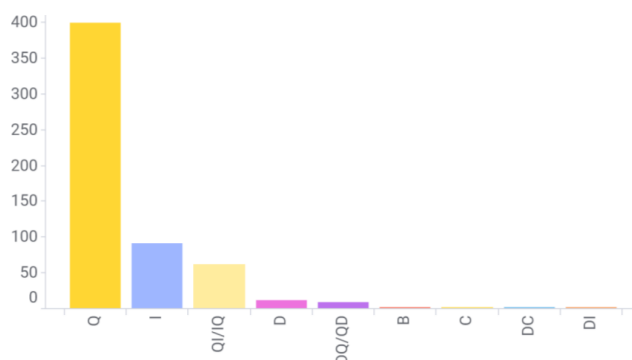


Figure 8—Bar chart showing the number of samples (y-axis) that are dominated by each cement type. Q: Quartz overgrowth, I: Illite, QI/IQ: Mixed quartz and illite, D: Dolomite cement, B: Baryte cement, C: Calcite cement, DC: Mixed dolomite and calcite, DI: Mixed dolomite and illite.

SEM imaging of the inter-granular volume show the abundant quartz cement in the form of quartz overgrowth (QO) whilst illite (I) dominating the authigenic clay minerals in the form of delicate micro-fibrous and leaf like crystalline fabric confirming the authigenic origin of this cementing mineral. The illite appears related to feldspar dissolution and commonly associated with sandstone with thin drapes of argillaceous laminae, whilst quartz cement appears to be very high in samples with massive appearance perhaps due to dewatering and subsequent removal of clay drapes. Additionally, the crystalline fabric of the illite has undoubtedly influenced the pore system due to the pore-bridging nature of the illite, effectively turning the pore network into poorly disconnected microporous system with very complex and tortuous pathway with a profound impact in the permeability for fluids to flow in the subsurface reservoir (Figure 9).

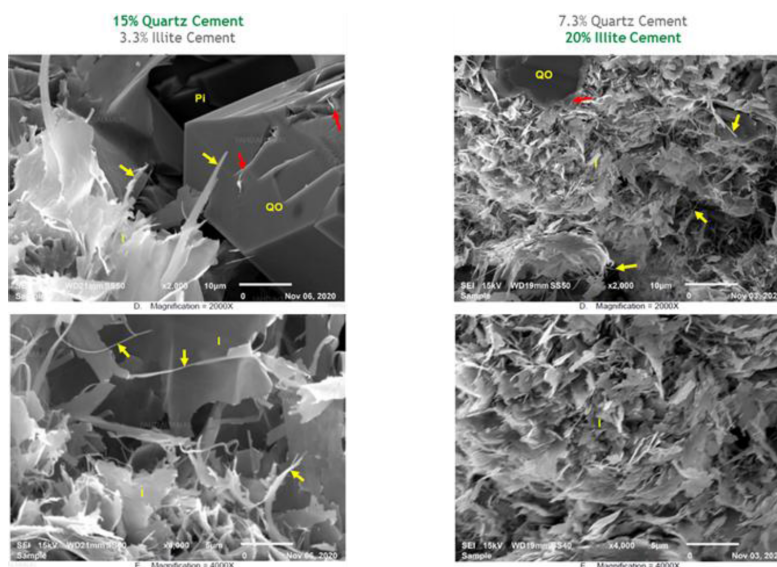


Figure 9—Left: two SEM images at different magnification levels highlighting sample with high quartz overgrowth. Right: SEM images of a sample dominated by authigenic pore-bridging and woven fibrous illite cement in the pore volume (Pi: intergranular pore, QO: quartz overgrowth, I: illite). Yellow arrows indicated the delicate fibrous micro-crystal of the illite.

Relationship between Reservoir Cementation with Pore Types

Quantitative evaluation of the pore types in such tight sandstones is essential and can be obtained from the point count data. Four rock types can be found in clastic reservoirs. Originally, sandstones are deposited with high intergranular pores. Then due to acidic fluid movement, certain labile grains such as feldspar tend to partially or completely dissolve creating secondary dissolution pores. The third type is common in fine-grained sediments and clastic reservoirs that go through diagenetic processes in which a network of small and poorly interconnected microporosity is developed that usually alter the original inter-granular porosity. The fourth is fracture porosity which is rare and mostly associated with the tectonic movement and structural deformation of rocks generating intense fractures that can add pore volume for fluids storage in what was once a very tight or non-porous reservoir. In this study, it has been found that most of the reservoir porosity is mainly contributed by a micropore system that is associated with facies that is in part fully dominated by authigenic illite which is present in the pore space as fibrous and pore-bridging crystalline form. In fact, the statistical data show that nearly 70% of the pore system is mainly derived from microporosity, 20% from dissolution porosity whilst the remaining 10% is the only contribution of the original inter-granular porosity. Such results are well correlated with the reservoir's poor to moderate sorting, moderate to high level of compaction, high level of authigenic quartz cementation and illite crystallization (Figure 10).

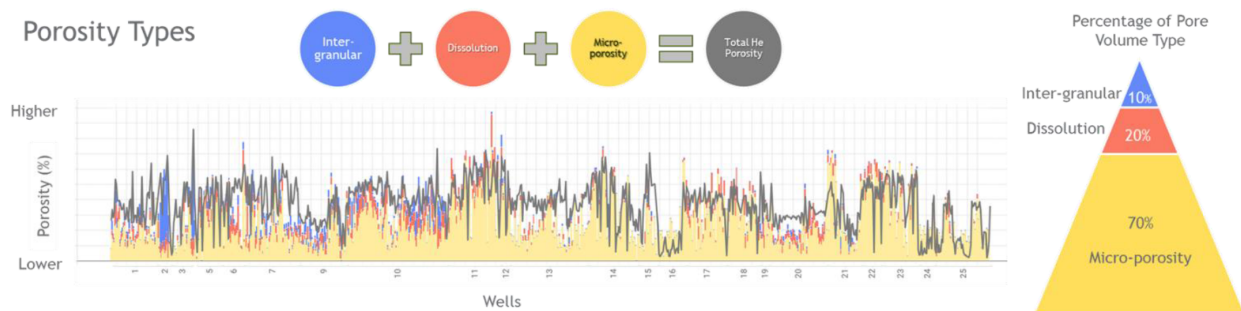


Figure 10—Left: Bar chart showing the contribution of porosity from each pore type for all samples in the study together with the plugs He measured porosity curve. The quantification of the three different types of pores is very much equal to the total plug porosity providing high confidence to the point count modal analysis of pore types. The color-coded bar charts clearly illustrate the dominance of micro-porosity in almost all wells except well 2. **Right:** a pyramid showing the average relative proportion of pore types in the entire reservoir.

The understanding of the dominance of micro-porosity in the subject reservoir and the relationship between cement type and facies has been the most important finding from the diagenetic and point count study with implications to reservoir characterization, classification and rock typing focusing on pore volume distribution in the reservoir facies as well as interpretations of dynamic flow data such as mercury injection capillary pressure analysis (MICP) and also geomechanical characterization of the reservoir. Integrating all the results of the diagenetic and core facies analysis, a classification scheme has been established which subdivided the reservoir into four main rock types, RT-1 to RT-4, with descending reservoir quality from highest to lowest. RT-1 describes a clean sandstone that is interlaminated with very thin clay drapes which are usually dominated by illite cement and measured the highest reservoir porosity. RT-2 is a sandstone which is usually very clean, dewatered and appears massive with no clay interlamination. This facies is usually dominated by quartz cement and subsequently measures lower porosity than RT-1. Sandstones that usually have moderate to high detrital clay content are lumped in the RT-3 class which are the lowest reservoir quality, whilst all non-reservoir facies such as very muddy and heterolithic facies are lumped in the RT-4 class (Table 1)

Table 1—A proposed classification scheme for a tight sandstone reservoir with strong diagenetic influence and a fairly homogenous quartz arenitic composition.

Rock Type Code (RT)	Facies Description	Dominant Material Impacting Pore Volume	Nature of Pore System	Reservoir Quality
RT1	Clean Stratified Sandstone	Pore Bridging Authigenic Illite	Dominated by Microporosity	Good Reservoir Quality
RT2	Clean Massive Sandstone	Pore Filling Quartz Overgrowth	Tightly Packed Intergranular Porosity	Moderate Reservoir Quality
RT3	Argillaceous Sandstone	Fine Grains and Detrital Clay Composition	Low Preserved Porosity in the Form of Micro to Nano Pores	Low Reservoir Quality
RT4	Non-Reservoir Lithologies	Detrital Clay and Pore Sorting of Heterolithic Facies	None	Non-Reservoir

Geomechanical Characterization

The UCS test is considered the most reliable method for assessing rock strength (ISRM, 2007), but it is a destructive, costly, and punctual test that only represents the strength of the rock at a particular depth. In contrast, the scratch test and MRT provide continuous profiles of UCS. The scratch test has been widely used to derive a continuous UCS profile (Huang et al., 2011), while the MRT provides a manual measurement of the rebound of a metallic ball on the core, from which static Young's modulus and UCS can be derived (Katz et al., 2000). To ensure data quality and validity, we validated the UCS values from scratch and MRT against uniaxial compressive test results. From all our extensive measurement, our sandstone is classified as strong to extremely strong, based on the UCS values obtained from the testing program (Figure 11).

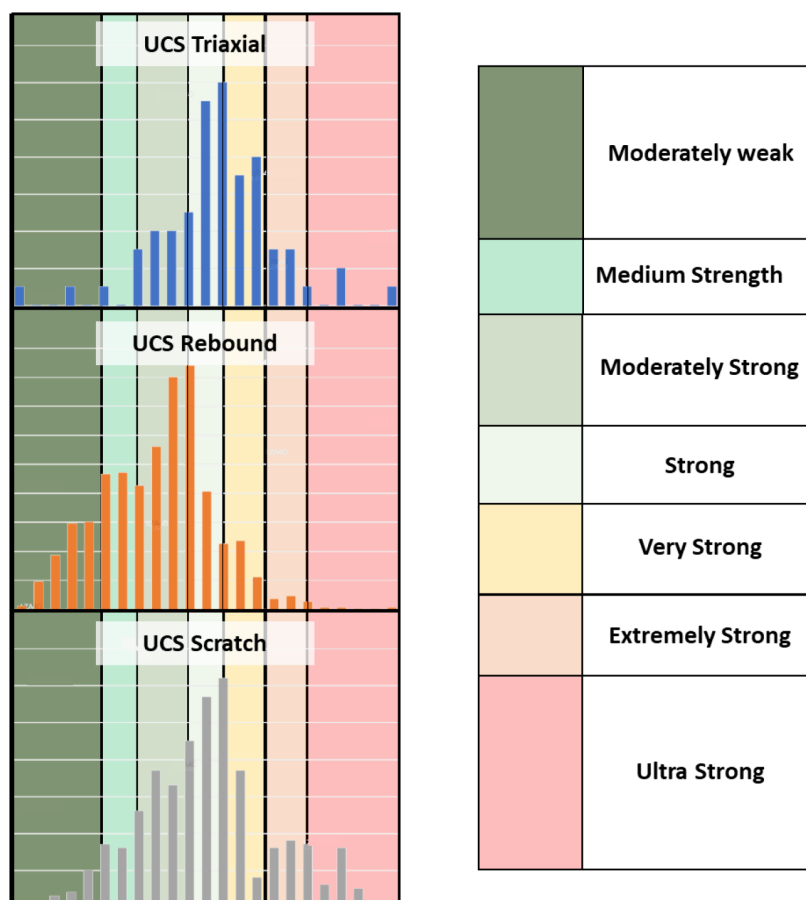


Figure 11—Histogram of UCS for triaxial test (0 confining pressure), rebound test and scratch test

An investigation into the elastic and strength anisotropy was conducted, and the results indicate that the tight sandstone exhibits isotropic behavior in terms of its mechanical parameters. This is evident from the overlap between horizontal and vertical sample measurements, as shown in Figure 12 (Left), suggesting that the rock's mechanical properties are consistent in all directions.

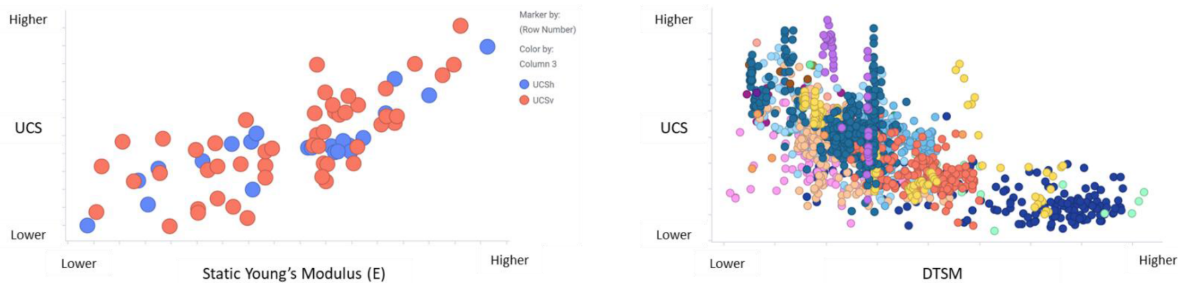


Figure 12—Left: UCS horizontal and vertical versus static Young's modulus for horizontal and vertical sample. Right: Shear slowness DTSM) versus UCS for triaxial, rebound and scratch data

Figure 13 presents an example of the reconstructed Uniaxial Compressive Strength (UCS) logs, alongside laboratory measurements of UCS, including triaxial tests at zero confining pressure, Micro Rebound Tests (MRT) using a Rebound Hammer, and scratch tests. The MRT and scratch tests provide detailed information that cannot be captured by sole triaxial testing, such as the variation of strength at the local scale. By integrating all the available data, it was possible to recompute a profile of strength that fairly represents the variation of strength of the tight sandstone, providing a more comprehensive understanding of its mechanical properties.

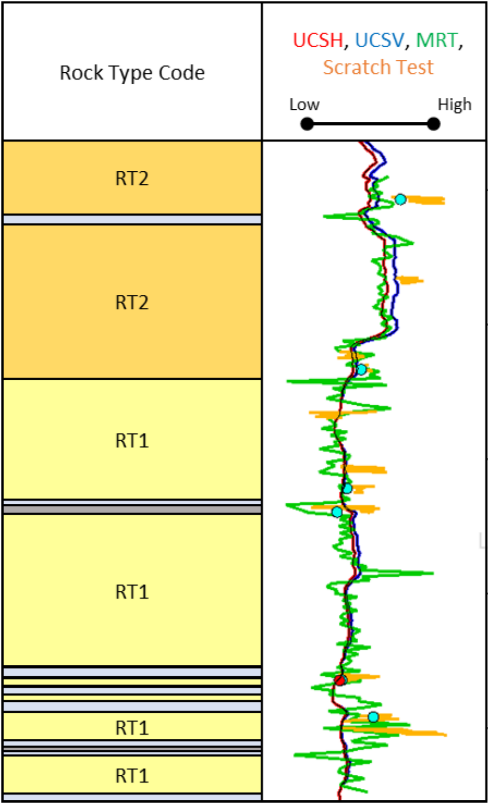


Figure 13—Example of a comparison of the UCS from a triaxial test at zero confining pressure (light blue and red disks) with MRT (green curve) and scratch (orange curve), plotted next to the identified rock types of the tight sandstone.

Linking Reservoir Diagenesis to Geomechanics Through Mechanical Rapid Micro-Rebound Testing (MRT)

A significant observation was made when conducting MRT analysis on a core from one of the wells. The core description documented the blocky, fine to medium grained sandstone of the tight reservoir interbedded with thin and soft sediment deformed argillaceous sandstone and siltstone units. Very slight difference in the clay intercalation was observed visually with the uppermost package showing cleaner sandstone with a massive appearance most likely due to dewatering. The middle and lower sections of the core showed some signs of dewatering, yet some thin clay intercalations were still preserved. This lithological difference is very minute to the extent that GR data measured on the core didn't show any considerable difference across the entire core.

The CCA He porosity data of the plugs provided an indication of lower porosity in the upper section most likely due to increased cementation. After conducting MRT analysis on the core, the uppermost package showed higher UCS values establishing a link between reservoir diagenesis particularly due to higher quartz overgrowth cementation in the massive sandstone facies (Figure 14). It is remarkable to note that such very subtle differences in the lithological composition (presence or absence of very thin layers of clay-drapes and laminations which have also been modified by rapid deposition causing varying levels of dewatering and clay re-distribution) have caused substantial differences in the reservoir quality to the extent that it even caused a difference in the geomechanical characteristics of this tight sandstone.

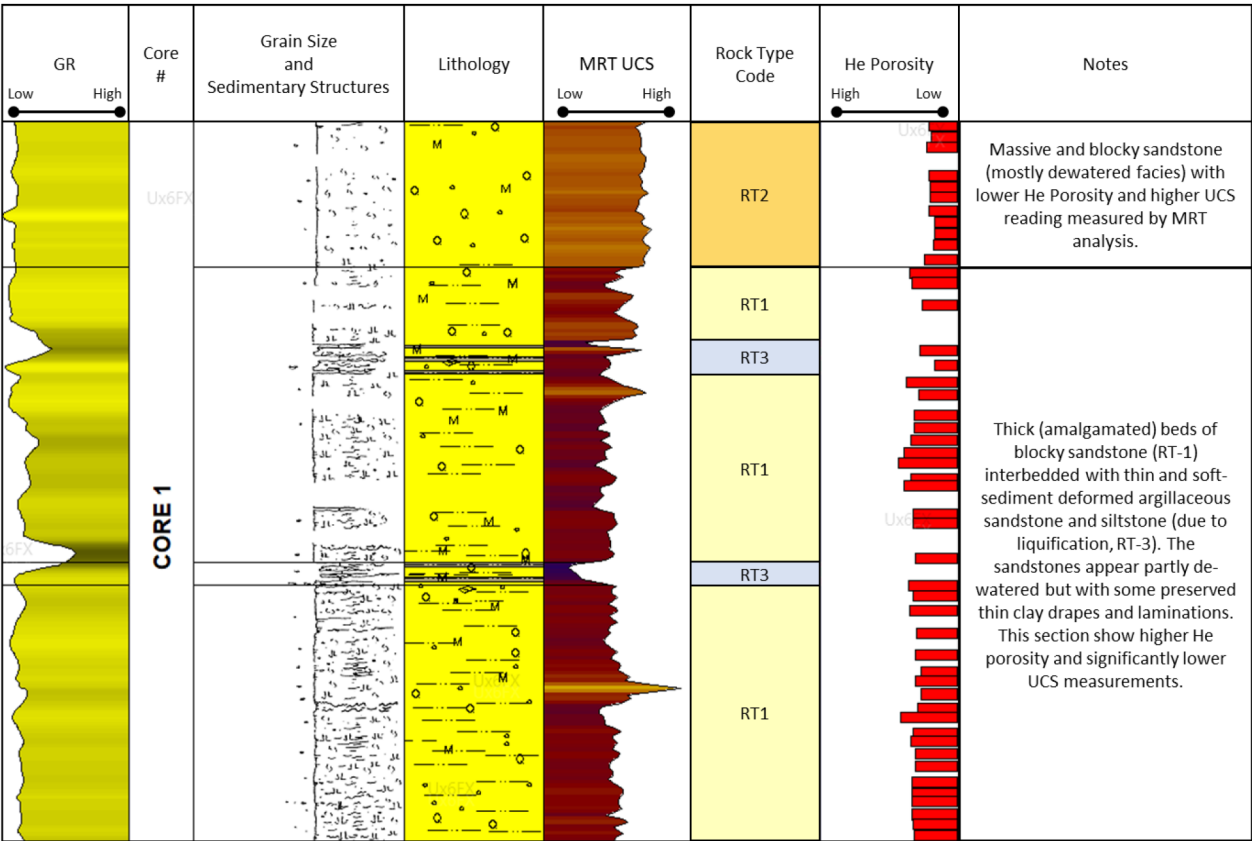


Figure 14—Core description chart showing core GR, grain size, sedimentary structure and lithology logs together with UCS and He Porosity data measured on cores by MRT and CCA analysis, respectively.

Relationships between UCS with Cement Types and Porosity

To confirm and validate the relationships between the dominant cement types and UCS of the tight sandstones, data from all core plugs color coded by rock types have been plotted for authigenic quartz and illite cements (Figures 15 and 16). Higher values of quartz cementation are positively correlated with

higher UCS data, whilst illite clay mineral cementation shows somehow a negative relationship. Meanwhile, porosity clearly increases with lower UCS highlighting the role of reservoir diagenesis and cementation to increase densification and rock hardness. Additionally, despite some overlaps between the different plugs, it can be seen clearly that the massive sandstone rock type (RT-2) generally is associated with lower porosity and higher UCS than RT-2 laminated sandstone facies. Additionally, another interesting finding is that the quartz cemented massive sandstones (RT-2) appear to have much narrower range of porosities (i.e. well-defined cluster) highlighting the effectiveness of quartz cement as a pore filling material homogenizing the reservoir into a very tight and geomechanically harder rock (i.e. higher UCS values, [Figure 17](#)).

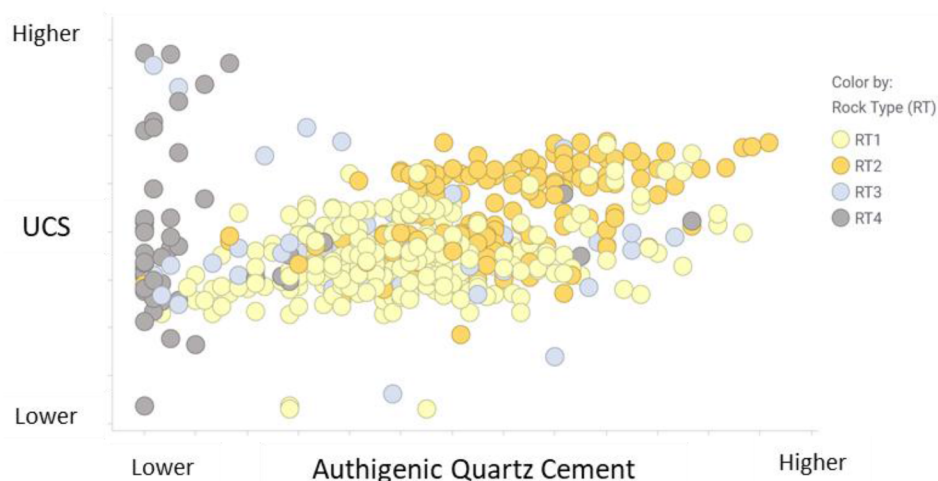


Figure 15—Plot showing the positive relationship between UCS and quartz overgrowth cement

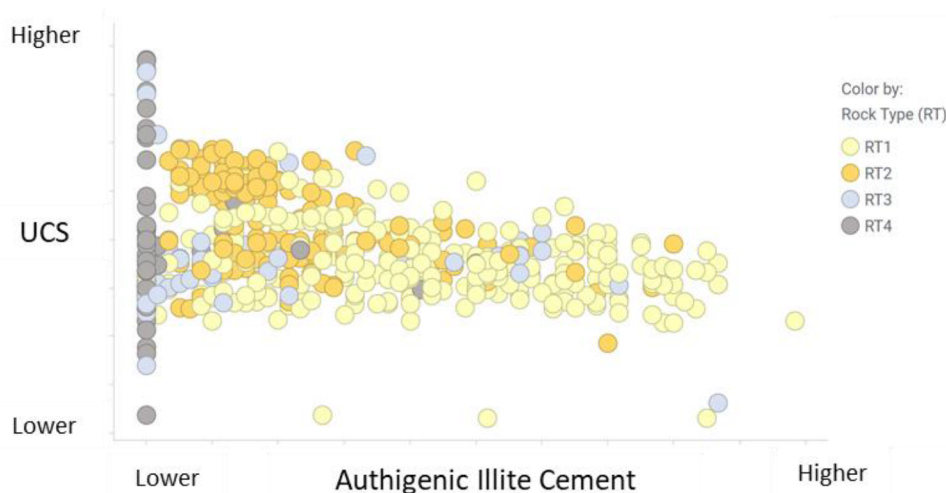


Figure 16—Plot showing the negative relationship between UCS and authigenic illite clay cement

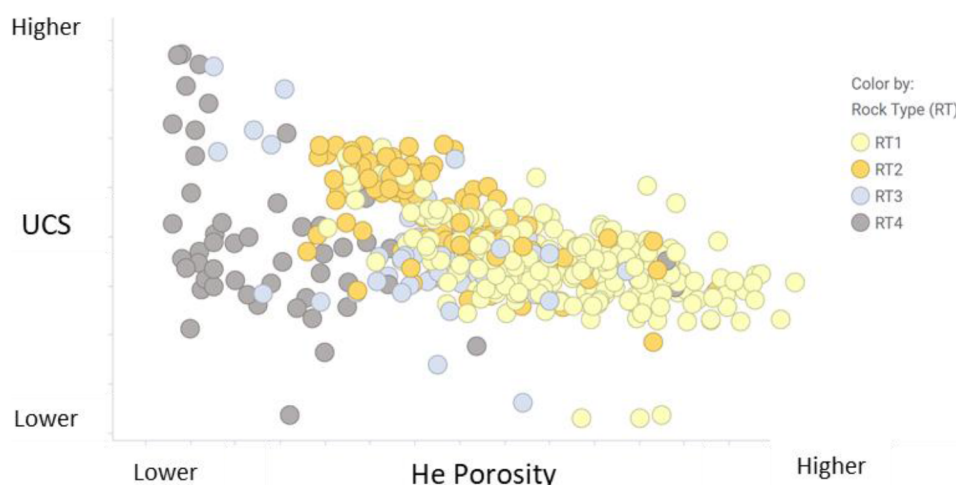


Figure 17—Plot showing the negative relationship between UCS and He Porosity. Rock types are color coded showing the lower porosities for massive sandstone (RT-2) whilst higher porosities for the laminated sandstone (RT-1). Notice how the RT-2 facies cluster in a very small zone with very high UCS data which is related very high quartz cementation. Actual values of UCS, He porosity and cementation data are proprietary and cannot be disclosed.

Conclusion

This study has provided a comprehensive characterization of a deeply buried tight sandstone reservoir, highlighting the intricate interplay between depositional characteristics of a fairly homogenous quartz arenitic sandstone with strong diagenetic processes that impacted reservoir quality. The reservoir's complex history of diagenetic evolution has led to a reduction in porosity and permeability, yet it has preserved a significant and dense network of poorly connected micropores present within the fibrous illite clay crystals. The study established a strong correlation between cement types and unconfined compressive strength (UCS). Quartz-cemented sandstones exhibit higher UCS values, indicating denser and harder rock, while illite-cemented sandstones show a negative relationship with UCS defining a relatively softer and more porous rocks that are ideal for trapping hydrocarbon which can be extracted through hydraulic fracturing of long horizontal wells. The relationships established between the two main reservoir rock types (RT-1 and RT-2) with UCS can be used to predict these facies and support reservoir's facies propagation for static geo-cellular models for accurate pore volume estimation and in-place resource calculation.

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